

The Regiment Sort and its Applications

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Abstract

This paper introduces Regiment Sort, a hierarchical sorting and aggregation operator in which an array is partitioned into equal-sized regiments, each regiment is sorted internally in ascending order, an aggregate score is computed for every regiment, and the regiments are subsequently concatenated in descending order of aggregate score. Structural padding is represented by a sentinel and therefore does not contribute to aggregate scores. Ties between regiment scores are resolved stably.

The construction is mathematically elementary but conceptually broad. It combines local ordering with global aggregation and can consequently be interpreted as a hierarchical ranking primitive for computer science, statistics, economics, finance, operations research, logistics, biology, network science, psychology, and abstract military resource allocation. Connections are discussed to order statistics, aggregation, portfolio ranking, clustering, scheduling, welfare economics, econometrics, machine learning, and decision theory. The paper emphasizes the distinction between the mathematical operator and any particular application domain.

The paper ends with “The End”

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1 Introduction

Many allocation problems possess a hierarchical structure. Individual objects belong to groups; objects may first be ordered within their groups, while the groups themselves must subsequently be ranked according to some aggregate measure of importance, strength, value, or utility.

Regiment Sort formalizes this structure.

Let an array contain MN positions. It is partitioned into r regiments, each containing mn positions, where

$$MN = rmn. \quad (1)$$

Each regiment is independently sorted in ascending order. Its aggregate score is then computed, with the default score being the arithmetic sum. Finally, the regiments are ordered in descending order of their scores and concatenated.

The resulting operation is therefore neither ordinary sorting nor ordinary aggregation. It is a two-level ordering mechanism:

$$\text{local ordering} \longrightarrow \text{group aggregation} \longrightarrow \text{global group ordering}. \quad (2)$$

The distinction is important. A conventional global sort discards the original group structure. Regiment Sort deliberately preserves that hierarchy.

The mathematical ideas underlying the construction are closely related to order statistics, aggregation operators, ranking theory, and hierarchical data structures [5, 8, 9]. Its computational implementation is straightforward, but its interpretation can be extended to many domains.

2 Definition of Regiment Sort

Let

$$A = (a_1, a_2, \dots, a_{MN}) \quad (3)$$

be the input array.

Partition A into r consecutive regiments,

$$R_i = (a_{(i-1)mn+1}, \dots, a_{imn}), \quad i = 1, \dots, r. \quad (4)$$

For each regiment, let

$$R_i^\uparrow = (a_{i,(1)}, a_{i,(2)}, \dots, a_{i,(mn)}) \quad (5)$$

denote its ascending order statistic representation, satisfying

$$a_{i,(1)} \leq a_{i,(2)} \leq \dots \leq a_{i,(mn)}. \quad (6)$$

The default regiment score is

$$S_i = \sum_{j=1}^{mn} a_{i,j}. \quad (7)$$

Let π be the permutation satisfying

$$S_{\pi(1)} \geq S_{\pi(2)} \geq \cdots \geq S_{\pi(r)}. \quad (8)$$

The Regiment Sort operator is

$$\text{RS}(A) = R_{\pi(1)}^\uparrow \parallel R_{\pi(2)}^\uparrow \parallel \cdots \parallel R_{\pi(r)}^\uparrow, \quad (9)$$

where $\parallel$ denotes concatenation.

Thus the algorithm has two distinct orderings:

elements are sorted within regiments, while regiments are sorted by aggregate strength.

(10)

3 Structural Padding and Sentinels

If the available data contain fewer than MN observations, padding may be required.

A numerical padding value is undesirable because it can contaminate the aggregate score. For example, padding by zero is innocuous only for a particular class of data and may be inappropriate when zero has substantive meaning.

We therefore introduce a structural sentinel

$$\mathbf{P}. \quad (11)$$

The sentinel is not an observation. It is a structural marker indicating an unoccupied position.

For a regiment R_i , define its substantive observations by

$$R_i^* = \{x \in R_i : x \neq \mathbf{P}\}. \quad (12)$$

Its score is consequently

$$S_i = \sum_{x \in R_i^*} x. \quad (13)$$

The sentinel is therefore excluded from both substantive ordering and aggregation. It may be retained structurally at the end of its regiment.

This distinction is important in statistical computation: missingness, structural absence, and a genuine numerical zero are not necessarily the same object.

4 Stable Regiment Ordering

Suppose two regiments satisfy

$$S_i = S_j. \quad (14)$$

Regiment Sort uses a stable ordering rule. If regiment i preceded regiment j in the original array, then

$$i < j \implies \pi^{-1}(i) < \pi^{-1}(j) \quad (15)$$

whenever their aggregate scores are equal.

This makes the operator deterministic and preserves information contained in the original ordering.

Stable sorting is particularly useful when the original order carries secondary information, such as chronological order, geographical order, priority class, or an externally determined identifier [13].

5 An Illustrative Example

Consider

$$A = (7, 1, 5, 2, 9, 3, 8, 4, 6, 10, 0, 11). \quad (16)$$

Let

$$M = 3, \quad N = 4, \quad m = 1, \quad n = 3. \quad (17)$$

There are therefore

$$r = \frac{MN}{mn} = \frac{12}{3} = 4 \quad (18)$$

regiments.

The original regiments are

$$R_1 = (7, 1, 5), \quad R_2 = (2, 9, 3), \quad (19)$$

$$R_3 = (8, 4, 6), \quad R_4 = (10, 0, 11). \quad (20)$$

Their scores are

$$S_1 = 13, \quad (21)$$

$$S_2 = 14, \quad (22)$$

$$S_3 = 18, \quad (23)$$

$$S_4 = 21. \quad (24)$$

Their local ascending orders are

$$R_1^\uparrow = (1, 5, 7), \quad (25)$$

$$R_2^\uparrow = (2, 3, 9), \quad (26)$$

$$R_3^\uparrow = (4, 6, 8), \quad (27)$$

$$R_4^\uparrow = (0, 10, 11). \quad (28)$$

Consequently,

$$\text{RS}(A) = (0, 10, 11, 4, 6, 8, 2, 3, 9, 1, 5, 7). \quad (29)$$

Notice that this is not the same as globally sorting A .

6 Pseudocode

The algorithm can be expressed independently of any programming language.

REGIMENT_SORT(A , M , N , m , n)

 verify $M*N$ is divisible by $m*n$

$r \leftarrow (M*N)/(m*n)$

 if $\text{length}(A) < M*N$:

 append structural sentinel P

 until $\text{length}(A) = M*N$

 for $i = 1, \dots, r$:

$R[i] \leftarrow$ i th regiment

 remove sentinel elements for substantive computation

 sort substantive elements of $R[i]$ ascending

 append sentinel elements structurally

$S[i] \leftarrow$ sum of substantive elements

 stably sort regiments by decreasing $S[i]$

 concatenate the sorted regiments

 return concatenated array

7 Python Implementation

The following implementation uses an internal sentinel object.

```
from typing import Any, List, Sequence, Tuple
```

```
class _PaddingSentinel:
```

```
    """Internal marker for structural padding."""
```

```
    __slots__ = ()
```

```
    def __repr__(self) -> str:
        return "PADDING"
```

```
PADDING = _PaddingSentinel()
```

```
def regiment_sort(
```

```

array: Sequence[Any],
M: int,
N: int,
m: int,
n: int,
) -> List[Any]:

    if M <= 0 or N <= 0 or m <= 0 or n <= 0:
        raise ValueError(
            "All dimensions must be positive."
        )

    total_size = M * N
    regiment_size = m * n

    if total_size % regiment_size != 0:
        raise ValueError(
            "M * N must be divisible by m * n."
        )

    if len(array) > total_size:
        raise ValueError(
            "Input exceeds the requested array size."
        )

    data = list(array)

    while len(data) < total_size:
        data.append(PADDING)

    number_of_regiments = (
        total_size // regiment_size
    )

    regiments = []

    for i in range(number_of_regiments):

        start = i * regiment_size
        end = start + regiment_size

        regiment = data[start:end]

        values = [
            x for x in regiment
            if x is not PADDING
        ]

        padding = [
            x for x in regiment
            if x is PADDING

```

```

    ]

    sorted_values = sorted(values)

    sorted_regiment = (
        sorted_values + padding
    )

    regiment_sum = sum(values)

    regiments.append(
        (
            i,
            sorted_regiment,
            regiment_sum
        )
    )

    # Python's sort is stable.
    regiments.sort(
        key=lambda x: -x[2]
    )

    result = []

    for _, regiment, _ in regiments:
        result.extend(regiment)

    return result

```

8 Correctness

8.1 Permutation Property

Assume first that there is no padding.

Proposition 1. Regiment Sort preserves the multiset of input observations.

Proof. Every regiment is transformed only by permutation through its internal ascending sort. The subsequent ordering of regiments also merely permutes whole regiments. Therefore every input observation occurs exactly once in the output. Hence the output is a permutation of the input. $\square$

With structural padding, the corresponding statement applies to the augmented array containing the sentinels.

8.2 Local Ordering

Proposition 2. Every substantive regiment in the output is internally sorted in ascending order.

Proof. The local transformation explicitly applies an ascending sort to every regiment before the global regiment permutation. Permuting complete regiments cannot change their internal ordering. $\square$

8.3 Global Ordering

Proposition 3. The regiment scores in the output are weakly decreasing.

Proof. The regiment permutation π is defined so that

$$S_{\pi(1)} \geq S_{\pi(2)} \geq \cdots \geq S_{\pi(r)}. \quad (30)$$

Concatenation therefore produces the required descending score sequence. $\square$

8.4 Stability

Proposition 4. If two regiments have equal scores, their original relative ordering is preserved.

Proof. The secondary key is absent. A stable sorting procedure therefore leaves equal-key elements in their original relative order. $\square$

9 Computational Complexity

Suppose there are r regiments of size mn .

Sorting each regiment independently using comparison sorting requires

$$O(mn \log(mn)) \quad (31)$$

operations per regiment, and hence

$$O(rmn \log(mn)) = O(MN \log(mn)). \quad (32)$$

Sorting the r regiment scores requires

$$O(r \log r). \quad (33)$$

Thus the overall comparison-based complexity is

$$O(MN \log(mn) + r \log r). \quad (34)$$

The principal computational distinction from global sorting is that the largest comparison operation is performed within groups rather than over the entire population.

This can be advantageous when groups have independent computational resources and can therefore be sorted in parallel.

10 Parallel Computing Interpretation

Regiment Sort possesses an immediate parallel decomposition.

The local transformations

$$R_i \longrightarrow (R_i^\uparrow, S_i) \quad (35)$$

are independent across i .

Consequently, one may write

$$\mathcal{F}(A) = [f(R_1), f(R_2), \dots, f(R_r)], \quad (36)$$

where

$$f(R_i) = (R_i^\uparrow, S_i). \quad (37)$$

The global operation is then a comparatively small sorting problem on the r aggregate scores.

This architecture resembles the general divide-and-conquer philosophy of parallel algorithms [4].

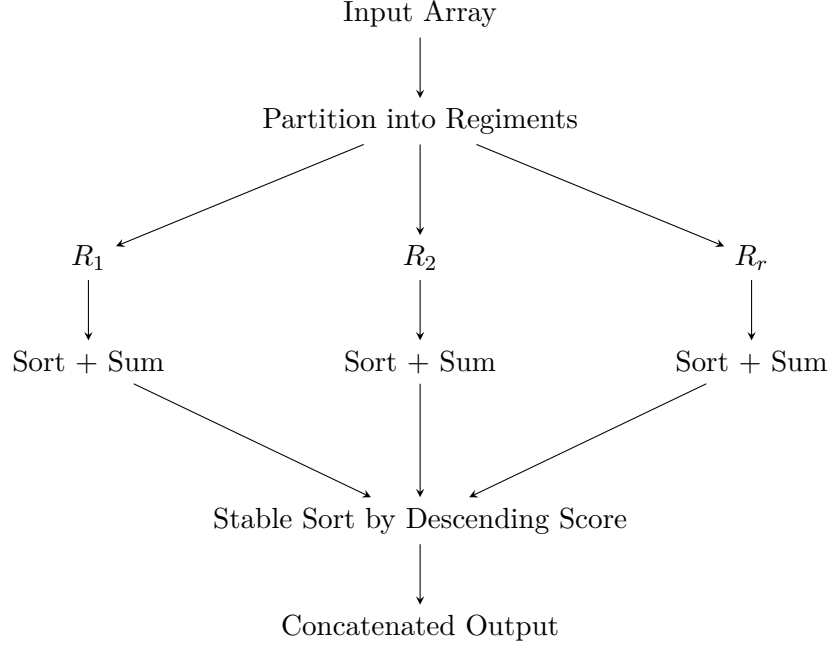


Figure 1: Sort architecture.

11 Generalized Regiment Scores

The sum is only one possible aggregation functional.

Let

$$S_i = G(R_i) \quad (38)$$

for a general functional G .

Possible choices include the mean,

$$G(R_i) = \frac{1}{|R_i^*|} \sum_{x \in R_i^*} x, \quad (39)$$

the maximum,

$$G(R_i) = \max R_i^*, \quad (40)$$

the minimum,

$$G(R_i) = \min R_i^*, \quad (41)$$

a weighted score,

$$G(R_i) = \sum_j w_{ij} x_{ij}, \quad (42)$$

or a nonlinear utility function,

$$G(R_i) = U(x_{i,1}, \dots, x_{i,mn}). \quad (43)$$

The generalized operator can therefore be written

$$\text{RS}_G(A) = R_{\pi(1)}^\uparrow \parallel \dots \parallel R_{\pi(r)}^\uparrow, \quad (44)$$

where π orders the values of $G(R_i)$.

This generalization connects Regiment Sort to multicriteria decision analysis, utility theory, and operations research [12, 22].

12 Statistics and Econometrics

From a statistical perspective, a regiment is a local sample and its score is an estimator or statistic derived from that sample.

For example, the sample mean is

$$\bar{X}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} X_{ij}. \quad (45)$$

Regiments can consequently be ranked according to estimates of location, dispersion, tail exposure, or other statistics.

Examples include

$$G_1(R_i) = \bar{X}_i, \quad (46)$$

$$G_2(R_i) = \text{median}(R_i), \quad (47)$$

$$G_3(R_i) = \text{Var}(R_i), \quad (48)$$

$$G_4(R_i) = \text{CVaR}_\alpha(R_i). \quad (49)$$

This allows Regiment Sort to operate as a data-analysis primitive in which observations are first normalized locally and populations are then ranked according to estimated characteristics.

In panel-data econometrics, a regiment could correspond to a country, firm, household, industry, or time window. The aggregate score could then represent an estimated economic quantity [25].

13 Finance

A natural financial application is portfolio or asset-group ranking.

Let R_i denote a portfolio containing asset returns r_{ij} . Its aggregate return is

$$S_i = \sum_j r_{ij}. \quad (50)$$

A risk-adjusted version may instead use

$$S_i = \frac{E[R_i] - r_f}{\sigma_i}, \quad (51)$$

which is structurally related to the Sharpe ratio [21].

A generalized Regiment Sort can therefore rank portfolios by performance, risk, liquidity, or a composite investment objective.

However, aggregate return alone does not constitute a complete portfolio decision rule. Diversification, covariance, drawdown, liquidity, and investor utility remain relevant [14, 15].

14 Economics and Welfare

Suppose regiment i represents a population group and x_{ij} represents an economic resource available to member j .

The sum

$$S_i = \sum_j x_{ij} \quad (52)$$

measures aggregate resources, but not necessarily welfare.

If social welfare is represented by an increasing concave utility function u , then

$$W_i = \sum_j u(x_{ij}) \quad (53)$$

may provide a more welfare-sensitive group score.

For example, diminishing marginal utility can be represented by

$$u'(x) > 0, \quad u''(x) < 0. \quad (54)$$

This demonstrates an important methodological distinction:

$$\boxed{\text{aggregate resources} \neq \text{aggregate welfare.}} \quad (55)$$

Regiment Sort is therefore best understood as a flexible ranking mechanism; the choice of score embodies the normative or positive assumptions of the application.

15 Operations Research and Logistics

In logistics, a regiment can represent a collection of tasks, warehouses, vehicles, machines, or supply channels.

Let

$$S_i = \sum_j q_{ij} \quad (56)$$

represent the quantity associated with group i .

Groups can then be prioritized by descending capacity.

A more general constrained optimization problem may be written as

$$\max_x \quad \sum_i U_i(x_i) \quad (57)$$

subject to

$$\sum_i c_i x_i \leq C, \quad (58)$$

$$x_i \geq 0. \quad (59)$$

Regiment Sort does not solve this optimization problem by itself. Rather, it provides a deterministic ranking primitive that may be embedded inside a larger allocation algorithm.

16 Computer Science and Data Structures

Regiment Sort is naturally interpreted as a hierarchical data-processing operation.

Its structure resembles grouped reduction followed by ranking:

$$\text{Group} \rightarrow \text{Map} \rightarrow \text{Reduce} \rightarrow \text{Sort} \rightarrow \text{Concatenate} . \quad (60)$$

This makes it compatible conceptually with map-reduce architectures [6].

The local computation is

$$R_i \mapsto (\text{sort}(R_i), G(R_i)) , \quad (61)$$

followed by the global reduction

$$\{(R_i, G(R_i))\}_{i=1}^r \mapsto \text{sort}_{\downarrow G} . \quad (62)$$

This separation can be exploited in distributed computing, streaming systems, database aggregation, and parallel data processing.

17 Artificial Intelligence and Machine Learning

In machine learning, a regiment may represent a batch, cluster, expert, agent population, or local environment.

Suppose a model produces observations

$$x_{ij} = f_{\theta}(z_{ij}) . \quad (63)$$

A group-level score can be constructed as

$$S_i = \frac{1}{n_i} \sum_j \ell(x_{ij}, y_{ij}) , \quad (64)$$

where ℓ is a loss function.

Regiment Sort can then rank groups according to their estimated loss, reward, uncertainty, or other model-derived statistic.

In reinforcement learning, one could define

$$S_i = \sum_{t=0}^T \gamma^t r_{it} , \quad (65)$$

where γ is a discount factor. The resulting ordering ranks groups according to cumulative reward.

Such uses should be distinguished from learning itself: Regiment Sort is an ordering mechanism, not a learning algorithm.

18 Biology and Ecology

Biological systems frequently exhibit hierarchical organization: individuals belong to populations, populations to communities, and communities to ecosystems.

A regiment can therefore represent a population and its score can represent population abundance, biomass, genetic diversity, or another measurable quantity.

For example,

$$B_i = \sum_j b_{ij} \quad (66)$$

may represent biomass in population group i .

Alternatively, a diversity score may be based on the Shannon index

$$H_i = - \sum_k p_{ik} \log p_{ik}, \quad (67)$$

where p_{ik} is the proportion of species k in group i [20].

Thus the same mathematical architecture can rank biological populations without assuming that the sum is necessarily the appropriate biological criterion.

19 Physics

The framework also admits a physical interpretation.

Suppose particles are partitioned into ensembles R_i , with an observable x_{ij} associated with particle j . The group score may be an extensive quantity,

$$S_i = \sum_j x_{ij}, \quad (68)$$

such as total mass or total energy.

Alternatively, an intensive statistic may be used, such as temperature,

$$T_i = \frac{1}{n_i} \sum_j T_{ij}. \quad (69)$$

This highlights the distinction between extensive and intensive variables, which is fundamental to statistical mechanics and thermodynamics [18, 3].

Regiment Sort itself is not a physical law. It is a computational representation that can organize observations generated by physical systems.

20 Chemistry

A similar construction can be used for chemical datasets.

A regiment may represent a molecular family, experimental batch, or reaction class. Its score may be an aggregate concentration, yield, activity, or estimated free-energy-related quantity.

For example, if y_{ij} denotes reaction yield,

$$S_i = \sum_j y_{ij} \quad (70)$$

ranks experimental groups according to total observed yield.

A more sophisticated score might combine multiple measurements,

$$S_i = w_1 Y_i + w_2 P_i - w_3 C_i, \quad (71)$$

where Y_i is yield, P_i is purity, and C_i is cost.

Again, the ranking mechanism is generic; scientific interpretation comes from the chosen variables and model.

21 Psychology, Psychiatry, and Human Decision Making

Human decision processes frequently involve hierarchical aggregation. Individuals may first evaluate alternatives locally and subsequently aggregate judgments at the group level.

Let x_{ij} represent an assessment by individual j within group i . A simple group score is

$$S_i = \sum_j x_{ij}. \quad (72)$$

A weighted model can account for differing reliability,

$$S_i = \sum_j w_{ij} x_{ij}, \quad w_{ij} \geq 0. \quad (73)$$

However, psychological phenomena such as group polarization, conformity, anchoring, framing effects, and bounded rationality imply that simple aggregation need not reproduce actual human judgment [11, 23].

Clinical psychiatry introduces an additional methodological caution: clinical states cannot legitimately be reduced to arbitrary numerical scores without validated measurement instruments and appropriate clinical interpretation. Regiment Sort can organize validated measurements, but it does not constitute a diagnostic or therapeutic method.

22 Philosophy and Logic

The operator also illustrates a general philosophical distinction between the properties of parts and the properties of wholes.

If

$$S_i = \sum_j x_{ij}, \quad (74)$$

then the group score is an aggregate property. But an aggregate property does not necessarily characterize every constituent.

This is the familiar methodological problem of aggregation.

In logical terms, the transformation has the form

$$\{x_{ij}\} \longrightarrow \{R_i\} \longrightarrow \{G(R_i)\} \longrightarrow \text{Order}(\{G(R_i)\}). \quad (75)$$

The operator therefore separates ontology at the element level from ordering at the group level.

This distinction is closely related to philosophical debates concerning parts and wholes, emergence, and reductionism [1].

23 Political Science and International Relations

A state can be represented abstractly as a multidimensional capability vector,

$$\mathbf{c}_i = (c_{i1}, c_{i2}, \dots, c_{ik}). \quad (76)$$

A scalar aggregate score could be

$$S_i = \mathbf{w}^T \mathbf{c}_i. \quad (77)$$

Regiment Sort can then rank states, organizations, or policy alternatives according to a declared index.

This can be useful analytically, but scalarization is inherently lossy. Two states may possess equal aggregate scores while having radically different capability profiles.

Consequently,

$$S_i = S_j \not\Rightarrow \mathbf{c}_i = \mathbf{c}_j. \quad (78)$$

The stable tie rule preserves ordering information but does not eliminate the underlying multidimensional distinction.

24 Diplomacy and Geopolitics

A diplomatic system can similarly be represented as a collection of relationships,

$$D_{ij} = f(E_{ij}, P_{ij}, S_{ij}, C_{ij}), \quad (79)$$

where the variables may represent economic, political, security, and cultural dimensions.

A group-level diplomatic score could then be used to organize negotiations or analytical priorities.

The crucial point is that Regiment Sort should be used as a descriptive ranking mechanism rather than as a substitute for diplomatic judgment. Diplomatic outcomes depend upon strategic interaction, incomplete information, credibility, institutions, bargaining power, and historical context [19, 16].

25 Warfare Economics and Abstract Military Allocation

The military metaphor motivating the name of the algorithm is naturally understood as a resource-allocation abstraction.

Suppose a collection of units forms a regiment, and each unit has a capability score x_{ij} . Then

$$S_i = \sum_j x_{ij} \quad (80)$$

represents aggregate nominal capability.

Regiment Sort would therefore rank formations by aggregate capability.

However, actual military effectiveness cannot generally be represented by a simple additive function. A more realistic abstract model could be

$$S_i = F(L_i, R_i, T_i, I_i, C_i, M_i), \quad (81)$$

where the variables represent logistics, readiness, terrain, information, coordination, and mission compatibility.

In particular,

$$\text{nominal strength} \neq \text{effective capability}. \quad (82)$$

This distinction is central to operations research and military economics. Regiment Sort can therefore be viewed as a computational primitive for ranking abstract formations or resource pools, rather than as a complete battlefield decision procedure.

26 Information, Uncertainty, and Robust Regiment Sort

Suppose the observed score is uncertain:

$$\hat{S}_i = S_i + \varepsilon_i, \quad (83)$$

where ε_i represents measurement error.

Then the ranking permutation π itself becomes uncertain.

A probabilistic formulation asks for

$$\Pr(S_i > S_j \mid \mathcal{D}), \quad (84)$$

where $\mathcal{D}$ denotes the available data.

A robust Regiment Sort could rank groups according to conservative estimates,

$$S_i^L = \hat{S}_i - z_\alpha \hat{\sigma}_i. \quad (85)$$

This connects the framework to statistical decision theory and robust optimization [24, 2].

27 Multiobjective Regiment Sort

A single sum may be inadequate when several objectives matter.

Let

$$\mathbf{S}_i = (S_{i1}, S_{i2}, \dots, S_{ik}). \quad (86)$$

One possibility is scalarization,

$$S_i = \sum_{k=1}^K w_k S_{ik}. \quad (87)$$

Another is Pareto ordering. Regiment i dominates regiment j if

$$S_{ik} \geq S_{jk} \quad \text{for all } k \quad (88)$$

and

$$S_{ik} > S_{jk} \quad \text{for at least one } k. \quad (89)$$

This produces a partial rather than total ordering [17, 7].

Thus the classical Regiment Sort is best regarded as the scalar special case of a more general hierarchical multiobjective ranking architecture.

28 Comparison with Ordinary Sorting

Table 1: Comparison with ordinary sorting

Property	Global Sort	Regiment Sort
Grouping	Discarded	Preserved
Local ordering	Global	Within regiment
Global criterion	Individual value	Regiment score
Aggregation	None	Explicit
Tie handling	Optional	Stable by definition
Padding	Usually external	Structural sentinel
Parallelism	Less naturally grouped	Natural local decomposition
Interpretation	Total element ranking	Hierarchical ranking

The distinction is therefore conceptual as well as computational.

29 General Theoretical Structure

Regiment Sort may be represented as the composition

$$\text{RS} = \text{Concat} \circ \text{Rank}_G \circ \text{LocalSort}. \quad (90)$$

More explicitly,

$$A \xrightarrow{\text{Partition}} (R_1, \dots, R_r) \xrightarrow{\text{LocalSort}} (R_1^\uparrow, \dots, R_r^\uparrow) \xrightarrow{G} (S_1, \dots, S_r) \xrightarrow{\text{Rank}} \pi \xrightarrow{\text{Concat}} \text{RS}(A). \quad (91)$$

This decomposition makes the construction modular.

One may change:

1. the partition rule;
2. the local ordering rule;
3. the aggregation functional;
4. the global ordering rule;
5. the treatment of missing or structural observations.

The resulting family of algorithms may be denoted

$$\text{RS}[P, L, G, O, H], \quad (92)$$

where P is partitioning, L is local ordering, G is aggregation, O is global ordering, and H is the handling of structural values.

30 Economic Interpretation

The algorithm has a particularly natural economic interpretation as a two-stage allocation mechanism.

At the first stage, resources are organized within groups. At the second stage, groups compete for priority according to an aggregate criterion.

Let x_{ij} represent resources assigned to agent j in group i . The group’s total endowment is

$$E_i = \sum_j x_{ij}. \quad (93)$$

Ranking groups by E_i produces a capacity hierarchy.
But if utility is nonlinear,

$$U_i = \sum_j u(x_{ij}), \quad (94)$$

then the ranking may differ.

Thus Regiment Sort makes explicit a central economic principle:

the ranking of groups depends on the aggregator used to define “strength.”

(95)

This is analogous to the broader problem of choosing social welfare functions, production functions, risk measures, and performance indices.

31 Limitations

Regiment Sort has several mathematical and practical limitations.

First, the sum is sensitive to scale. A group containing many observations can dominate a group containing fewer observations even when their means are comparable.

Second, sums can conceal distributional structure. Two regiments may have

$$S_i = S_j \quad (96)$$

while possessing very different internal distributions.

Third, scalarization can obscure multidimensional capability.

Fourth, measurement error can change the ranking.

Fifth, padding must be explicitly distinguished from genuine observations.

Finally, a ranking algorithm is not an optimization theory by itself. Choosing the order in which objects should be considered does not determine the optimal allocation of resources.

32 Research Extensions

Several extensions follow naturally.

32.1 Weighted Regiment Sort

$$S_i = \sum_j w_{ij} x_{ij}. \quad (97)$$

32.2 Risk-Adjusted Regiment Sort

$$S_i = \frac{\mu_i - r_f}{\sigma_i}. \quad (98)$$

32.3 Robust Regiment Sort

$$S_i = \inf_{\theta \in \Theta_i} G(R_i; \theta). \quad (99)$$

32.4 Bayesian Regiment Sort

$$S_i = E[G(R_i) \mid \mathcal{D}]. \quad (100)$$

32.5 Dynamic Regiment Sort

If the data evolve over time,

$$S_i(t) = G(R_i(t)), \quad (101)$$

and the ordering becomes

$$\pi_t = \text{Rank}(S_1(t), \dots, S_r(t)). \quad (102)$$

This creates a dynamic ranking system.

32.6 Hierarchical Regiment Sort

Regiments may themselves be grouped into larger formations:

$$R_{ij} \longrightarrow R_i \longrightarrow F_k \longrightarrow \mathcal{A}. \quad (103)$$

The procedure can consequently be recursively applied.

This produces a hierarchy analogous to trees, organizational structures, multilevel statistical models, and hierarchical clustering [10].

33 Conclusion

Regiment Sort is a simple algorithm with a broad mathematical interpretation.

Its fundamental operation is

$$\boxed{\text{sort locally} \rightarrow \text{aggregate locally} \rightarrow \text{rank globally} \rightarrow \text{concatenate.}} \quad (104)$$

The stable ordering rule provides deterministic treatment of equal scores, while structural sentinels permit arrays to be padded without contaminating the underlying statistics.

The algorithm is not a replacement for optimization, statistical inference, economic theory, military analysis, or machine learning. Its significance lies instead in providing a reusable computational abstraction that can be embedded within those disciplines.

At its most general level, Regiment Sort formalizes the principle that a complex population may be organized simultaneously at two scales:

$$\boxed{\text{order the parts, aggregate the groups, and order the groups.}} \quad (105)$$

This makes Regiment Sort a useful primitive for hierarchical computation, resource allocation, statistical organization, portfolio analysis, distributed computing, and multidimensional decision systems.

References

- [1] Aristotle. *The Complete Works of Aristotle*. Princeton University Press, 1995.
- [2] A. Ben-Tal, L. El Ghaoui, and A. Nemirovski. *Robust Optimization*. Princeton University Press, 2009.
- [3] H. B. Callen. *Thermodynamics and an Introduction to Thermostatistics*. Wiley, 1985.

- [4] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein. *Introduction to Algorithms*. MIT Press, 4th edition, 2022.
- [5] H. A. David. *Order Statistics*. Wiley, 1969.
- [6] J. Dean and S. Ghemawat. MapReduce: Simplified data processing on large clusters. *Communications of the ACM*, 51(1):107–113, 2008.
- [7] K. Deb. *Multi-Objective Optimization Using Evolutionary Algorithms*. Wiley, 2002.
- [8] C. Gini. Variability and mutability. *Studi Economico-Giuridici*, 1912.
- [9] W. Härdle and L. Simar. *Applied Multivariate Statistical Analysis*. Springer, 2007.
- [10] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning*. Springer, 2nd edition, 2009.
- [11] D. Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, 2011.
- [12] R. L. Keeney and H. Raiffa. *Decisions with Multiple Objectives*. Cambridge University Press, 1993.
- [13] D. E. Knuth. *The Art of Computer Programming, Volume 3: Sorting and Searching*. Addison-Wesley, 2nd edition, 1998.
- [14] H. Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.
- [15] R. C. Merton. An intertemporal capital asset pricing model. *Econometrica*, 41(5):867–887, 1973.
- [16] H. J. Morgenthau. *Politics Among Nations*. Alfred A. Knopf, 1948.
- [17] V. Pareto. *Manuale di Economia Politica*. Società Editrice Libreria, 1906.
- [18] F. Reif. *Fundamentals of Statistical and Thermal Physics*. McGraw-Hill, 1965.
- [19] T. C. Schelling. *The Strategy of Conflict*. Harvard University Press, 1960.
- [20] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948.
- [21] W. F. Sharpe. Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3):425–442, 1964.
- [22] H. A. Simon. A behavioral model of rational choice. *The Quarterly Journal of Economics*, 69(1):99–118, 1955.
- [23] A. Tversky and D. Kahneman. Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157):1124–1131, 1974.
- [24] A. Wald. *Statistical Decision Functions*. Wiley, 1950.
- [25] J. M. Wooldridge. *Econometric Analysis of Cross Section and Panel Data*. MIT Press, 2nd edition, 2010.

Glossary

Aggregation

A mapping from a collection of observations to a group-level statistic.

Array

An ordered finite collection of elements.

Comparator

A rule used to determine the relative ordering of two objects.

Econometric panel

Data containing observations indexed by multiple dimensions, commonly entities and time.

Extensive variable

A physical quantity whose magnitude scales with system size.

Global ordering

The ordering of entire regiments according to their aggregate scores.

Local ordering

The ordering of observations within an individual regiment.

Padding

Additional structural positions introduced to reach the required array dimension.

Pareto dominance

A relation in which one alternative is at least as good as another in every criterion and strictly better in at least one.

Regiment

A fixed-size subset of the input array.

Regiment score

The scalar quantity used to rank a regiment.

Regiment Sort

The hierarchical sorting operator defined in this paper.

Sentinel

A special structural marker representing the absence of a substantive observation.

Stable sorting

Sorting in which objects having equal keys retain their original relative ordering.

Structural padding

Padding that represents array capacity rather than an actual observation.

Scalarization

The transformation of a multidimensional criterion vector into a scalar criterion.

Utility

A numerical representation of preference or value.

The End